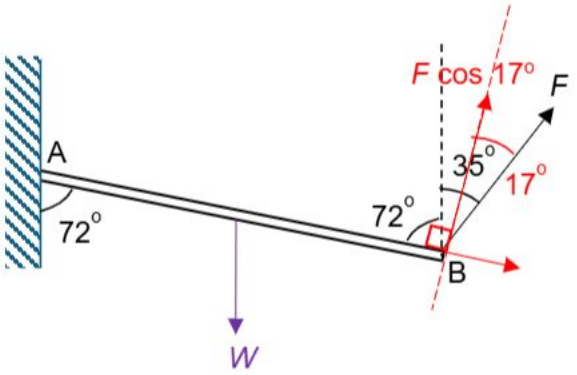
Question 1

(a)

Let the length AB be L.



$72^o+35^o-90^o=17^o$

Taking moments about A:

Anti-clockwise moment due to component of F perpendicular to AB = Clockwise moment due to weight

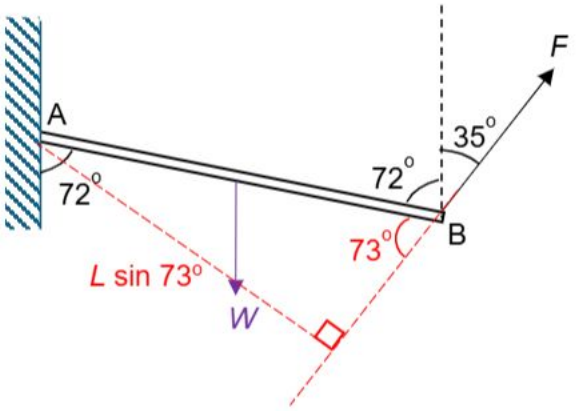
$$F\cos 17^o\,(L) = W\left(0.5L\sin 72^o\right)$$

$$F = 5.967 = 6.0 \text{ N}$$

B1

A0

Alternative:

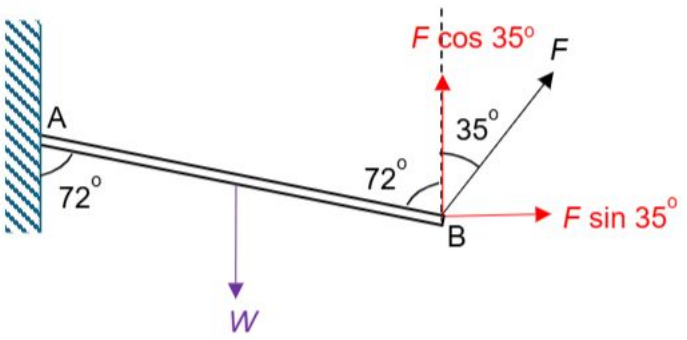
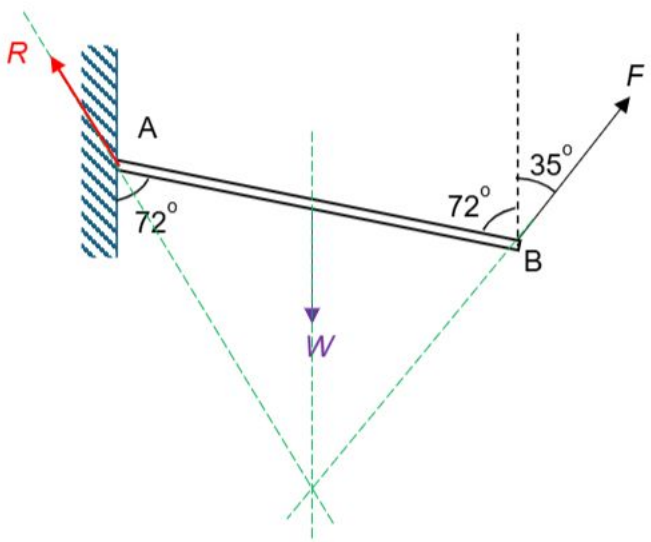


Taking moments about A:

Anti-clockwise moment due to F = Clockwise moment due to weight

$$F\left(L\sin 73^o\right) = W\left(0.5L\sin 72^o\right)$$

$$F = 5.967 = 6.0 \text{ N}$$

	Alternative:  Taking moments about A: Anti-clockwise moment due to F cos 35° + anti-clockwise moment due to F sin 35° = Clockwise moment due to weight $F\cos 35^o\left(L\sin 72^o\right)+F\sin 35^o\left(L\cos 72^o\right)=W\left(0.5L\sin 72^o\right)$ $F = 5.967 = 6.0$ N	
(b)(i)	 Note to marker: Ignore length of arrow, working lines are expected	**A1**
(b)(ii)	For equilibrium, $\sum F_x = 0$ $R_x = F_x = 6.0\sin 35^o$ $\sum F_y = 0$ $R_y + F_y = W$ $R_y = 12 - 6.0\cos 35^o$ $R = \sqrt{R_x^2 + R_y^2} = 7.89 = 7.9$ N Note to marker: Substituting unrounded F gives the same final answer (2 s.f.).	**B1** **B1** **A1**